



A Pangenome Sequence Search and Coordinate Translation Service for the UCSC Genome Browser

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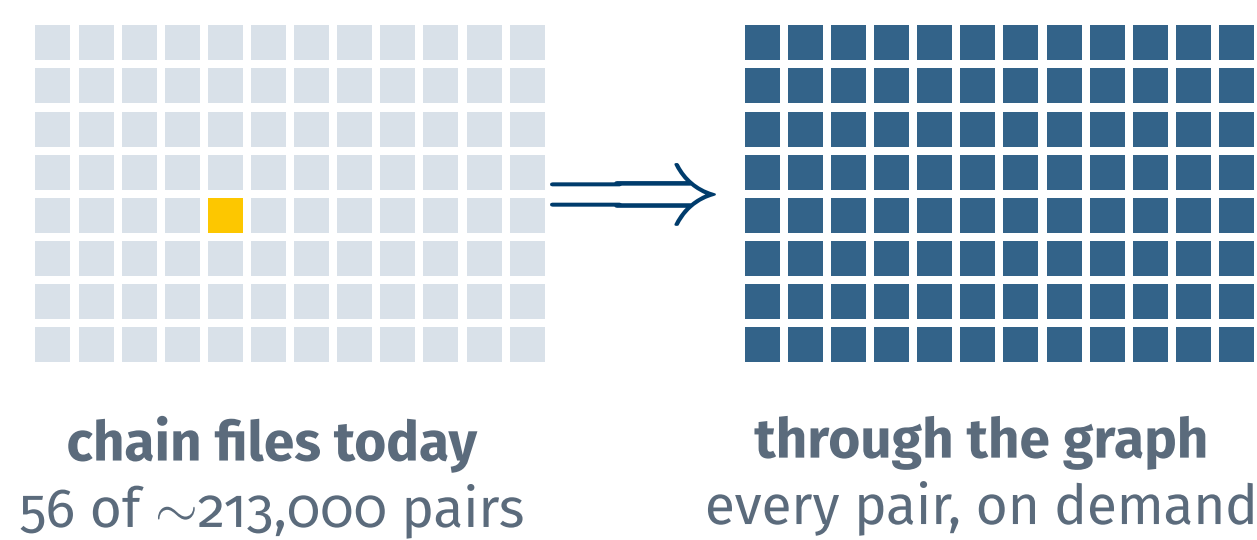


poster sources
parsaeskandar/hprc_2026_poster

Paste a **sequence** or a **position** — get an answer from **every HPRC haplotype**, live, inside the browser.

01 The graph nobody could ask

The HPRC v2.0 minigraph-Cactus graph holds **233 samples / 464 haplotypes**, yet there has been no interactive way to query it. Liftover needs a chain file per assembly pair — and almost none exist:



A researcher with a **sequence** can't ask which haplotypes carry it; one with a **position** can't move it between assemblies. We built the service that answers both, and wired it into the UCSC Genome Browser.

02 What users get



Pangenome sequence search

A BLAT-like page: a pasted sequence is mapped to the whole graph with `vg giraffe`; results rank all 464 haplotypes by identity and link to its position on any of them, drawn as a track with base-level mismatch coloring.



Any-to-any coordinate translation

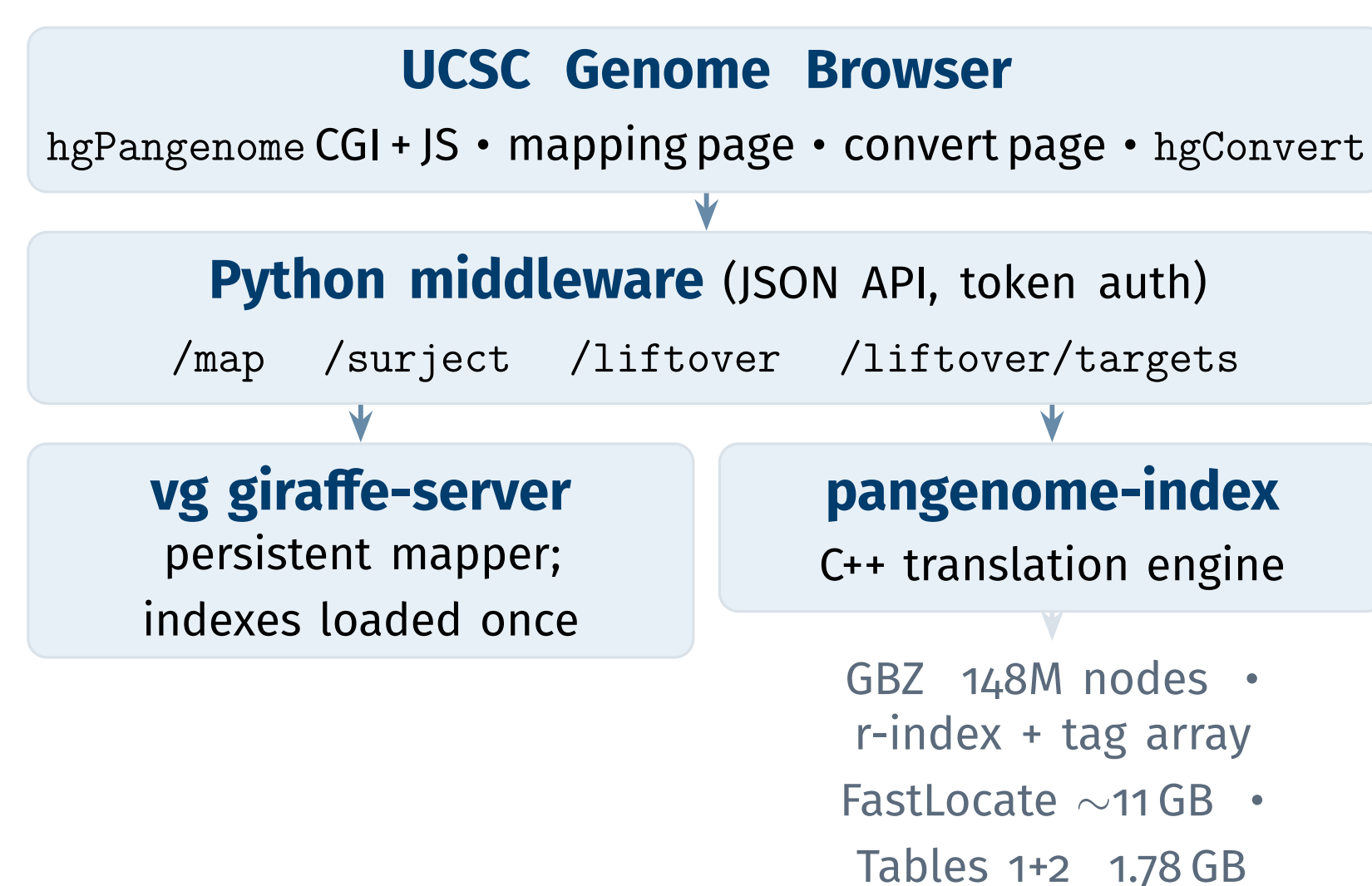
Liftover between any two of 466 assemblies at interactive speed, with a scored “which assemblies contain this region” destination picker.



Annotation lifting

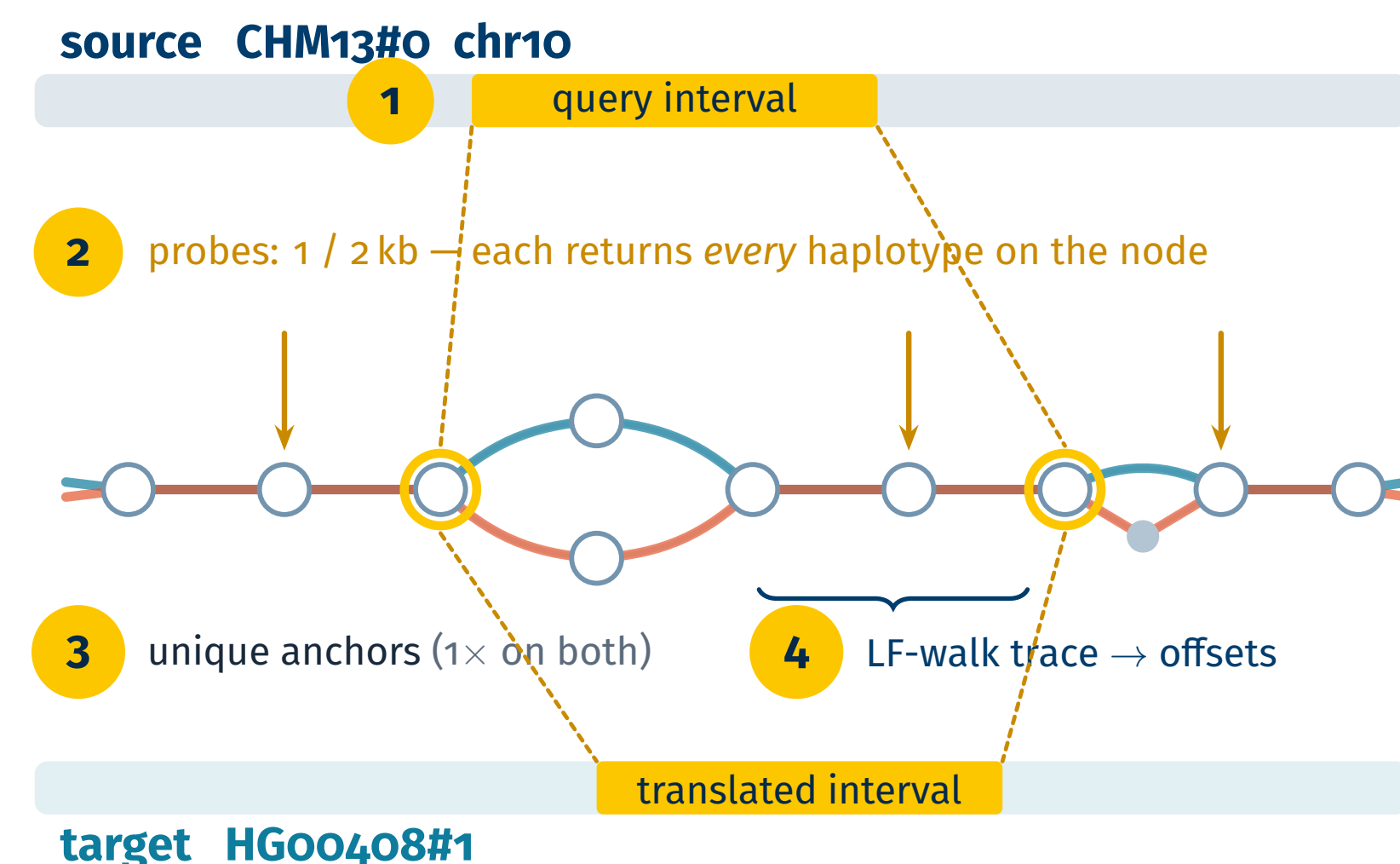
A QuickLift chain generated *at request time*: users land on an HPRC assembly with CHM13's genes, RefSeq, CenSat and T2T Encode at translated positions, plus an **Alignment Differences** track.

03 Architecture



Core browser changes: **3 files, 15 lines**. Everything else is a new, self-contained CGI — dark unless configured.

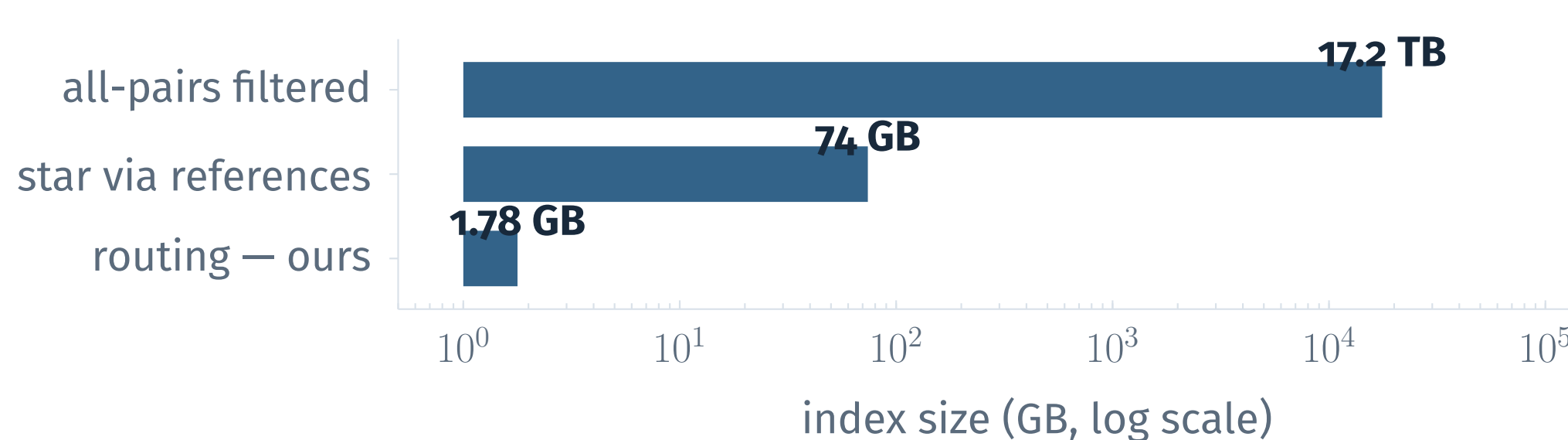
04 Translation through the graph



1 Table 1 turns `contig:start-end` into a graph path interval. 2 Probing discovers target candidates — or the **routing Table 2** skips it with a direct lookup. 3 Nodes visited exactly once by both sides are guaranteed orthologous; a colinearity gate drops implausible pairs. 4 The trace folds into **chain-semantics blocks**: broken by indels, flips, contig changes — never by substitutions.

05 A routing table that fits

An all-pairs interval table over the frequency-filtered graph (224M path fragments, $2,367\times$ fragmentation) would cost **17.2 TB**. Building on the *non-filtered* graph (94,554 paths) and storing *routing only* — which targets a source range touches, never their coordinates:

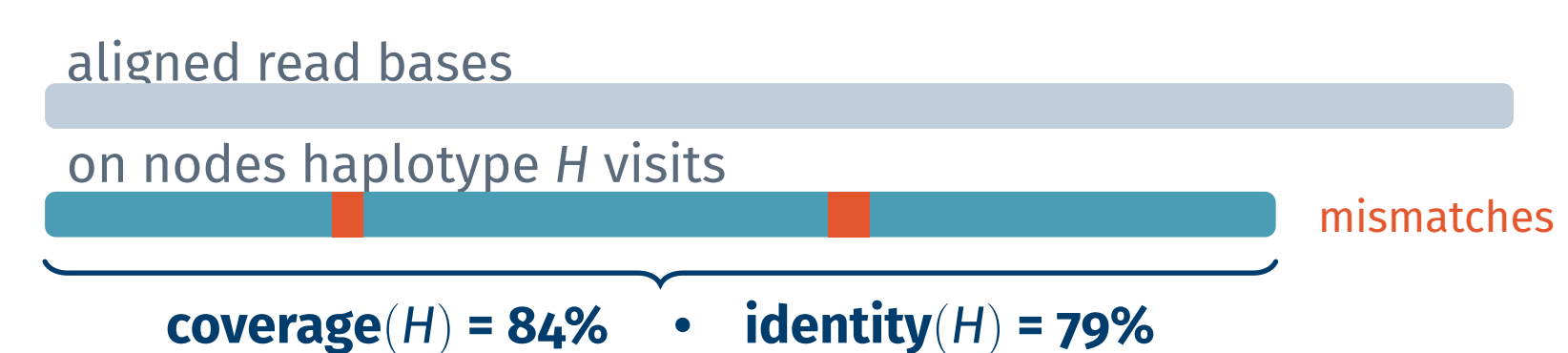


Dropping stored target intervals also removed every failure mode of interval pairing:

o abandoned runs (was 5.0M) o splits (was 35.7M) build: 35 min

06 Scoring all 464 at once

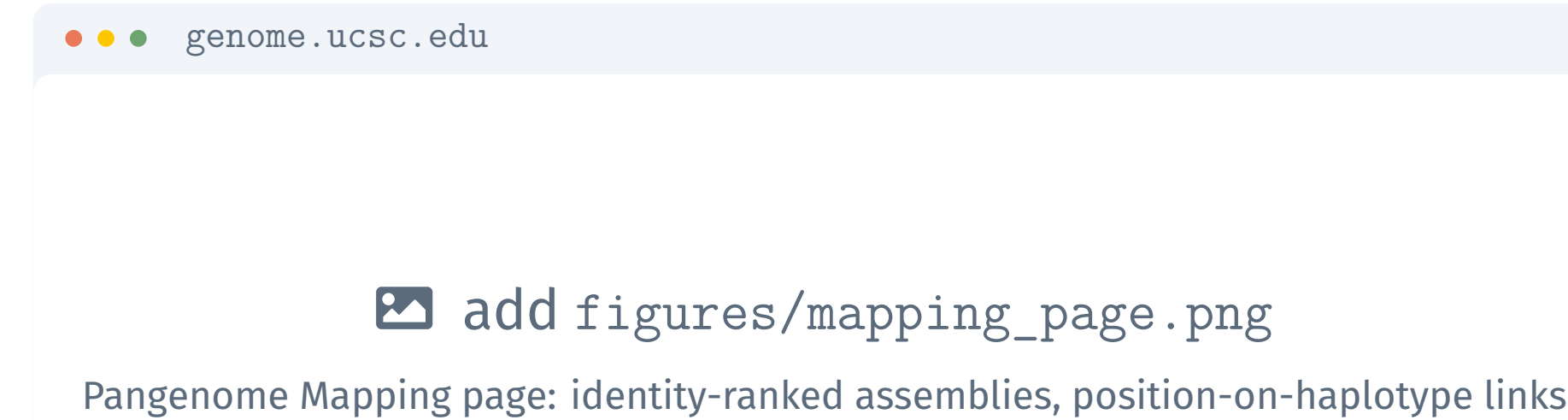
One pass over a mapped read's nodes scores **every haplotype** simultaneously — each probe already returns all of them:



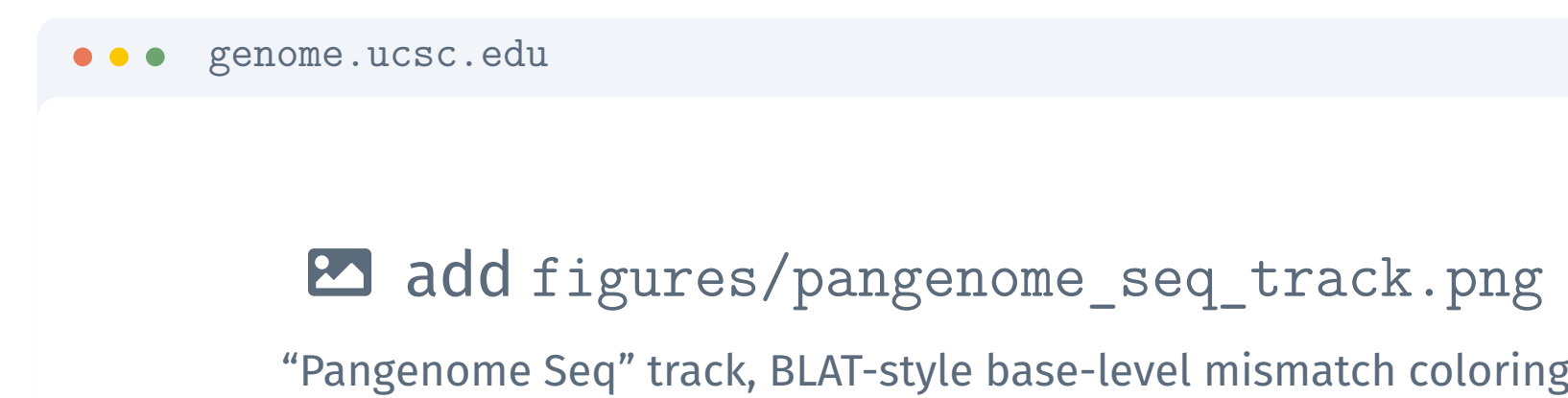
identity ≤ coverage always; the gap is the mismatch burden. Together they separate cases one number conflates:

100 / 99 present, near-exact 100 / 60 present, diverged
50 / 50 half missing

07 Sequence search, in the browser



A 300 bp query scores all 464 haplotypes in seconds. Picking a haplotype **re-surjects the cached alignment** — no remapping — and the browser opens there with the sequence as a native track:

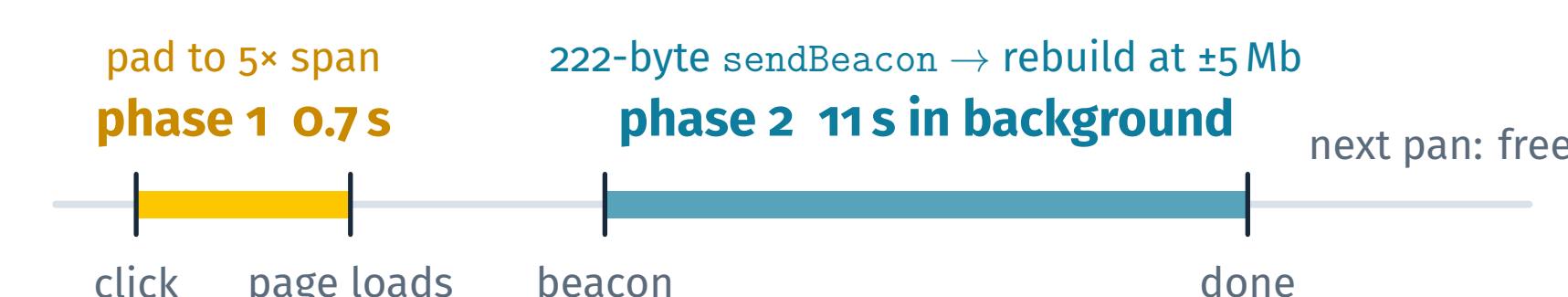


08 Landing with annotations

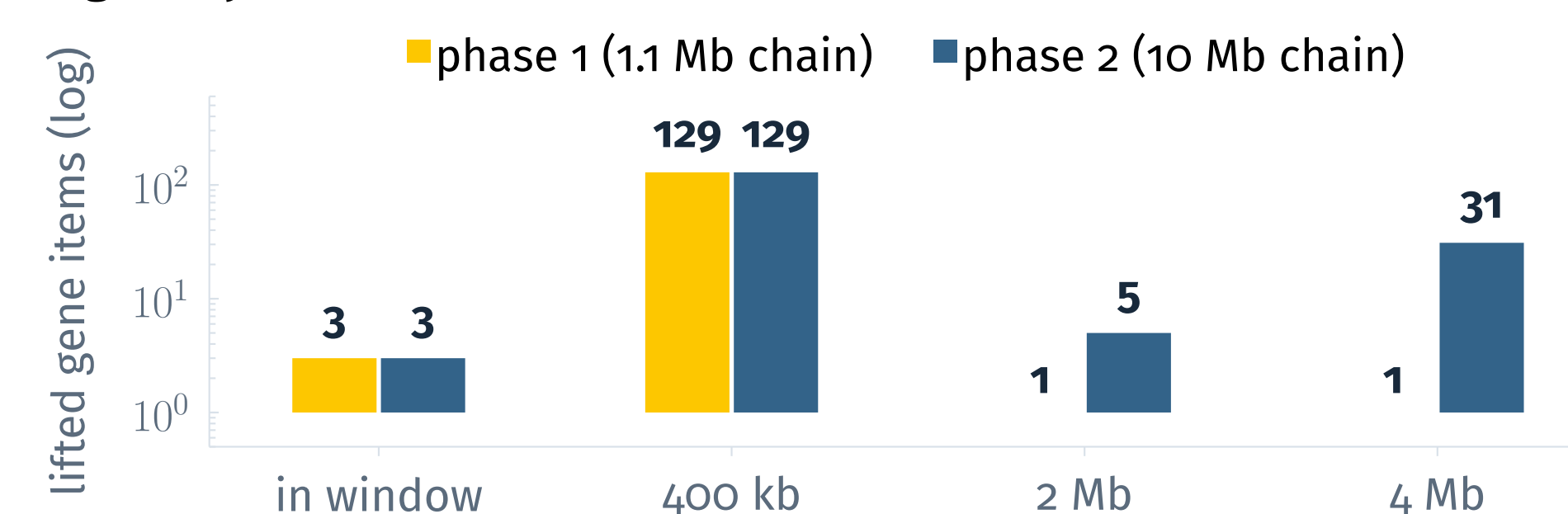


Converting a position lands on the target with **CHM13's annotation tracks lifted** through an on-the-fly chain, plus an **Alignment Differences** track — verified base-by-base: across 22 kb of ABO it shows exactly the 2 real C→T mismatches between hs1 and HG00597#1. Pangenome assemblies also appear in the stock hgConvert dropdown.

09 Chains that keep up with panning

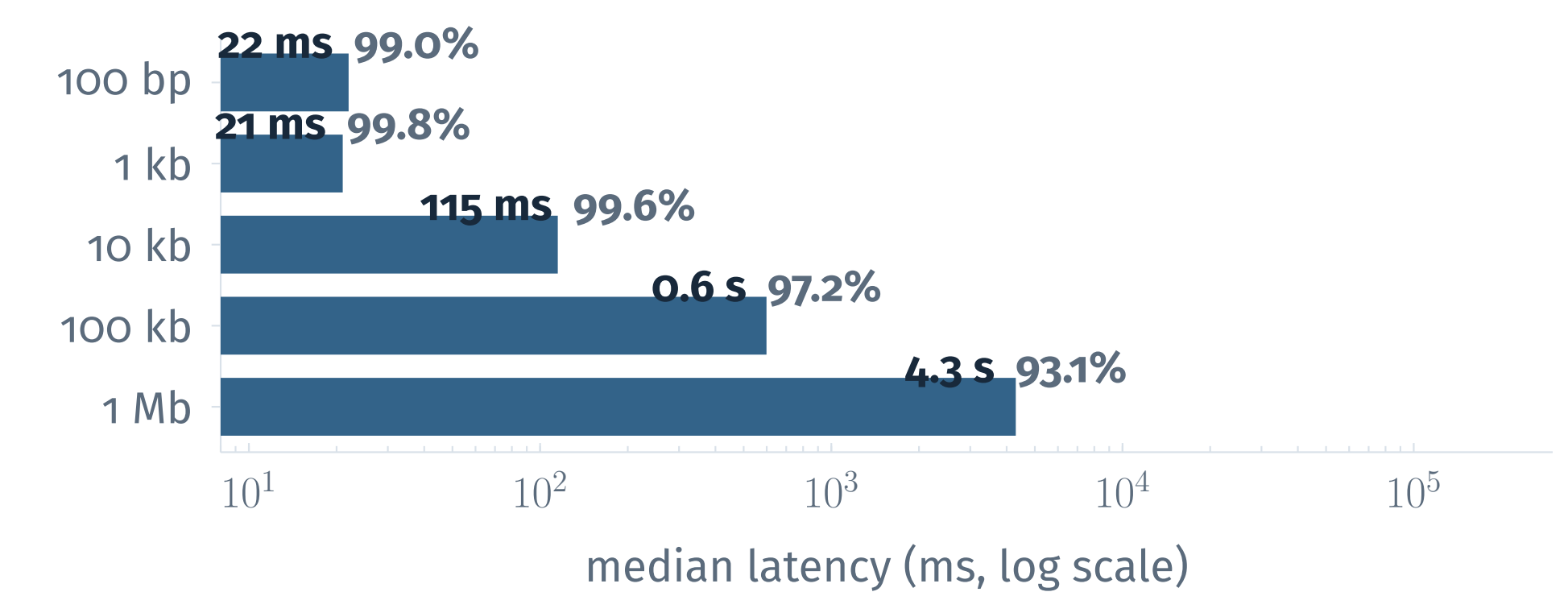


Blocks become bigChain+bigLink (JSON → BED → bedToBigBed); QuickLift reopens the file each request, so the widened chain appears on the next pan — *no reload*. Lifted items visible after panning away:



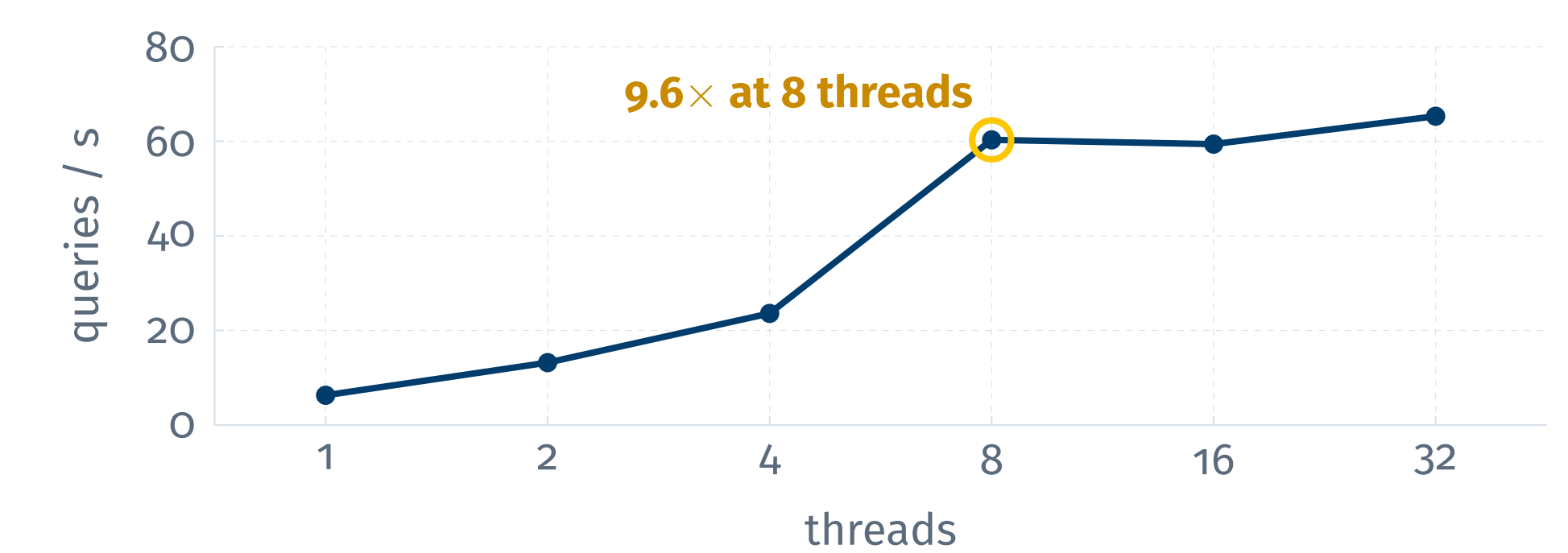
10 Performance

Translation latency (warm; completeness in gray):



Exhaustive candidate discovery (was stop-at-first-hit) took 1 Mb completeness **2.9%** → **93.1%** — and made it 2× faster.

Concurrent queries (3.4 kb, GIL released in C++):



Scored destination picker: 100 kb region in **1.28 s** vs 29.5 s exact (23×). Chain generation: linear, ~1.14 s/Mb.

11 What's next



Bulk offline chain generator

Direct path intersection instead of r-index probing: est. ~300× faster (~1.6 h per target genome-wide vs 19 days via the API) — precomputed whole-chromosome chains, full QuickLift parity.



Persistent chain cache

Widened chains currently die with the one-hour session; a cache makes popular regions free after the first visit.



Segdup-dense intervals

Wide queries lose positions (~46% at 1q21.1); exact path intersection removes the probe spacing that misses them.



Public release

Core-change sign-off, per-install API tokens, service-side concurrency deployment.

12 Acknowledgements

Built on `vg`, `GBWT/GBZ` and the tag-array index (Sirén et al.), the HPRC v2.0 minigraph-Cactus graph, and the UCSC Genome Browser (kent). `vg giraffe-server` is under review as `vgteam/vg PR #4879`. We thank the UCSC Genome Browser team for design feedback, and the Human Pangenome Reference Consortium.